



Home to over a million residents, Dharavi is being documented in unprecedented detail.

Narrow lanes, mixed land use, informal infrastructure and intense population density create challenges that conventional surveys often struggle to capture comprehensively. The digital twin offers planners, engineers and government agencies a common visual platform from which to understand existing conditions, coordinate interventions and plan future infrastructure with greater precision.



More Than a Planning Tool

Globally, digital twins are often associated with future-facing planning. Singapore uses them to simulate infrastructure and climate scenarios while Heathrow Airport has employed them to improve passenger movement and operational efficiency. European cities have used the technology to study energy use, urban comfort and climate resilience. Dharavi's application is different because the challenge is not simply planning a city of the future, but understanding and reorganising an existing urban ecosystem of extraordinary density and complexity.

Officials associated with the project describe the platform as a "single source of truth" for one of India's most ambitious urban transformation initiatives. Built using aerial drone imagery, ground-level LiDAR scans, lane surveys and digital door-to-door mapping, the model enables planners to study rehabilitation requirements, utilities, roads, drainage systems and public infrastructure before implementation begins.

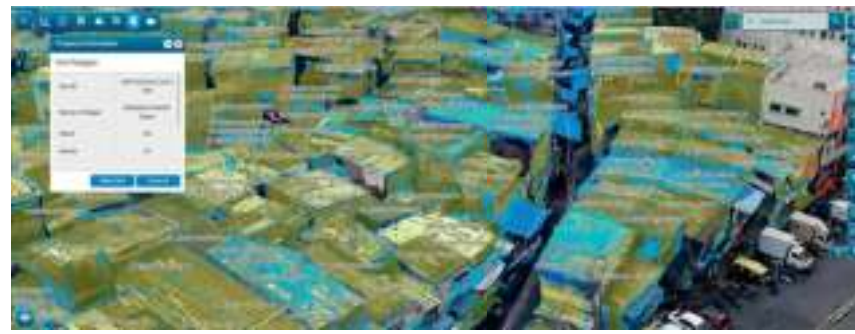
For Mumbai, a city that routinely grapples with flooding, transport disruptions and mounting climate pressures, the implications could eventually extend beyond redevelopment. Detailed digital mapping may support resilience planning, emergency response coordination and infrastructure management, creating a more informed framework for future urban decision-making.

Preserving the Past, Shaping the Future

Yet, perhaps the most compelling aspect of the exercise is that the digital twin is not merely a model of the future. It is also becoming an archive of the past.

As redevelopment transforms the landscape over the coming years, the virtual model will preserve a detailed record of the settlement's physical and social fabric. The lanes, rooftops, workspaces and neighbourhoods that have defined the area for generations will continue to exist in digital form long after the skyline changes.

For a project expected to transform the lives of more than a million residents, that may prove to be one of its most enduring legacies. The virtual city being created today is not simply helping build a new one. It is ensuring that the story of the old one is never lost.



A virtual Dharavi is taking shape, guiding planners with greater precision.

What is a Digital Twin?

A digital twin is a virtual replica of a physical asset or place that stays connected to real-world data and evolves as conditions change.

» Where did it originate?

The concept was first used by NASA in the early 2000s to monitor spacecraft systems and predict failures before they occurred.

» How is it used globally?

Today, digital twins are used for airports, factories, transport systems, buildings and entire cities to improve planning, execution and operations.

» What technologies does it involve?

Digital twins bring together drones, digital maps, sensors, cameras, cloud computing and AI-powered analytics to create a virtual replica of a real-world environment.

» What about India?

Digital twins are increasingly being used in smart-city programmes, infrastructure projects and large-scale urban redevelopment efforts such as Dharavi.

The Blueprint Before the Build

- Covers 621 acres of one of the world's densest slum clusters
- Maps over one lakh slum units
- Digitally captures lanes, buildings, infrastructure and open spaces
- Built using drone surveys, LiDAR mapping, lane recce and digital door-to-door visits
- Serves as a single source of truth for planning and execution
- Enables digital-first design review and construction monitoring
- Lays the foundation for long-term smart-city operations